CSC 281
Fibonacci Using Recursion and Iteration

Objective
In this assignment, you will implement the Fibonacci sequence in two different ways:
1. Recursive implementation
2. Iterative implementation using a loop

The goal is to compare performance and understand when to use each approach. All implementations must be written in Java.

Reminder: Fibonacci Definition
fib(n) = fib(n-1) + fib(n-2)
Base cases: fib(0)=0, fib(1)=1

Part 1: Implement Fibonacci in Two Ways
A. Recursive version
B. Iterative version

Note:

- You must write both the recursive and iterative versions in Java.
- Use standard Java libraries only.
- Do not use built-in functions or external libraries that compute Fibonacci for you.

Part 2: Compute Fibonacci Values
Test at least: n = 5, 10, 15, 20, 25, 30, 35, 40
Also attempt fib(100) using both methods.

Part 3: Measure Running Time
Record running time for each method and value of n.
Include a table with:
- n
- Recursive result
- Recursive time
- Iterative result
- Iterative time

Part 4: Create a Line Chart

Create a chart with:
- x-axis: n
- y-axis: running time
- one line for recursion
- one line for iteration

Part 5: Answer Questions

- Did both implementations produce the same results?

- Which implementation was faster? Why?

- How did recursive time change as n increased?

- How did iterative time change as n increased?

- Could you compute fib(100) recursively?

- Which implementation is better for large n and why?

- What repeated work occurs?

Submission Requirements

1. Source code
2. Report (with table, chart, answers)
3. Screenshot of program output